# Mohammad Kamran Ikram

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## Research Profile

PhD candidate at Swinburne University of Technology researching how electric vehicles and peer-to-peer energy sharing can support grid stability in the transition to net zero. Industry and applied research experience as a Grid Connection Engineer at Siemens (PSS® E studies for a 250 MW battery) and as a Research Fellow at CSIRO (power system restoration with grid-forming and grid-following inverters). Author of 5 journal articles, including a 2026 cover article in World Electric Vehicle Journal, 3 conference papers, 2 book chapters and 1 published patent.

## Education

**PhD, Electrical Engineering** Apr 2024 – 2027 (expected)
*Swinburne University of Technology, Melbourne, Australia*
**Research:** Peer-to-peer energy sharing and electric vehicle control, modelled in PSS® E. Fully funded by the Siemens–Swinburne Energy Transition Hub.

**Master of Technology, Instrumentation and Control Engineering** 2018 – 2020
*Aligarh Muslim University, India | GPA 8.69 / 10*
**Thesis:** Development of a reactive power transducer for static VAr compensator (SVC) applications.

**Bachelor of Technology, Electrical Engineering** 2013 – 2017
*Aligarh Muslim University, India*

## Research Interests

EV integration and charging optimisation · Peer-to-peer energy sharing · Power system restoration and black start · Grid-forming and grid-following inverters · Smart grids and microgrids · Battery energy storage grid studies · Real and reactive power measurement

## Research and Professional Experience

**Research Fellow** Nov 2025 – Jul 2026
*CSIRO, Melbourne, Australia | AR-PST Stage 5: System Restoration and Black Start*
- Evaluated how grid-forming (GFM) and grid-following (GFL) inverters can work together to replace synchronous generators in System Restart Ancillary Services (SRAS).
- Findings published in *Energies* (2026) on the coordinated operation and stability limits of GFM/GFL inverters in power system restoration.

**Research Assistant** Jul 2025 – present
*Swinburne University of Technology, Melbourne, Australia*
- Supporting research on EV participation, peer-to-peer energy trading and smart grid optimisation.

**Grid Connection Engineer** Feb 2025 – May 2025
*Siemens | Siemens–Swinburne Energy Transition Hub, Melbourne, Australia*
- Carried out generator performance standard (GPS) studies and grid connection assessments for a 250 MW battery energy storage system, including steady-state and dynamic analysis of the South Australian network in PSS® E.
- Integrated releasable user guide (RUG) models into OPDMS network data and maintained .dyr dynamic model files for system-wide contingency and dynamic studies.
- Wrote Python scripts that automate compliance checks against dynamic performance standards in single-machine infinite bus (SMIB) and NEM-wide studies.

**PhD Candidate** Apr 2024 – present
*Swinburne University of Technology, Melbourne, Australia*
- Developed a two-stage, fuzzy-informed optimisation framework that reduces peak load through hierarchical EV charging management (Journal of Energy Storage, 2025).
- Designed a state-of-charge-aware EV charging strategy and studied its effect on distribution network voltage (IEEE ETFG 2025), and the impact of charging station placement on hybrid microgrids (IEEE AUPEC 2025).

- Reviewed the role of EVs in peer-to-peer energy networks; selected as the cover article of World Electric Vehicle Journal, August 2026.

**Associate Researcher** Apr 2023 – Mar 2024

*Federation University Australia | Centre of Smart Analytics*

- Researched scheduling and operation of electric haul trucks in mining, and developed a benchmark dispatch model for an open-pit mine.

**Research Associate** Jan 2021 – Mar 2023

*Department of Electrical Engineering, Aligarh Muslim University, India | UGC DRS project, Government of India*

- Designed and experimentally validated transducer circuits and industrial prototypes for measuring real and reactive power, for voltage control and stability of smart grids and microgrids.
- Work led to two journal papers (IEEE Sensors Journal; Sensors and Actuators A) and a published patent.

## PUBLICATIONS

### Journal articles

**J1** **M. K. Ikram**, P. A. Hosseinabadi, M. Seyedmahmoudian, S. Mekhilef, and A. Stojcevski, "Coordinated operation and stability limits of grid-forming and grid-following inverters in power system restoration," *Energies*, vol. 19, no. 16, p. 3875, 2026. doi:10.3390/en19163875

**J2** **M. K. Ikram**, M. Seyedmahmoudian, G. Thirunavukkarasu, S. Mekhilef, A. Stojcevski, and J. Moreira, "A review of electric vehicle integration in peer-to-peer energy networks," *World Electric Vehicle Journal*, vol. 17, no. 8, p. 383, 2026. doi:10.3390/wevj17080383 **Cover article**

**J3** **M. K. Ikram**, M. Seyedmahmoudian, G. S. Thirunavukkarasu, S. Mekhilef, and A. Stojcevski, "A two-stage fuzzy-informed optimization framework for reduced peak load and hierarchical EV management in smart grids," *Journal of Energy Storage*, vol. 136, p. 118308, 2025. doi:10.1016/j.est.2025.118308

**J4** **M. K. Ikram**, M. S. J. Asghar, M. Seyedmahmoudian, S. Mekhilef, A. Stojcevski, and A. Al-Assaf, "Advanced real and reactive power measurement using analog multiplier and phase-controlled switching technique," *Sensors and Actuators A: Physical*, vol. 378, p. 115812, 2024. doi:10.1016/j.sna.2024.115812

**J5** **M. K. Ikram**, S. M. Amrr, M. S. J. Asghar, T. Islam, and A. Iqbal, "Voltage independent reactive current based sensor for static VAr control applications," *IEEE Sensors Journal*, vol. 23, no. 9, pp. 10023–10031, 2023. doi:10.1109/JSEN.2023.3258696

### Conference papers

**C1** **M. K. Ikram** et al., "Impact of an optimal SoC-aware EV charging strategy on the voltage regulation of distribution networks," in *Proc. IEEE Int. Conf. Energy Technologies for Future Grids (ETFG)*, Wollongong, Australia, Dec. 2025. doi:10.1109/ETFG61999.2025.11401024

**C2** **M. K. Ikram** et al., "Modeling and analysis of EV charging station placement impacts on a hybrid microgrid," in *Proc. IEEE Australasian Universities Power Engineering Conf. (AUPEC)*, Brisbane, Australia, 2025. doi:10.1109/AUPEC66173.2025.11219464

**C3** M. S. J. Asghar, **M. K. Ikram**, M. Sarfraz, and T. Islam, "A novel method for the measurement of active and reactive power by sample-and-hold," in *Proc. IEEE Int. Conf. Computing, Communication and Automation (ICCCA)*, Arad, Romania, 2021, pp. 424–428. doi:10.1109/ICCCA52192.2021.9666395

### Book chapters

**B1** S. Khan, M. Sufian, **M. K. Ikram**, and M. S. Khan, "A cloud-based innovation for establishing environmental surveillance: The smart air quality monitoring system," in *SIGMAA 2023*, Springer, vol. 1460, 2024. doi:10.1007/978-981-97-6349-8_20

**B2** **M. K. Ikram**, K. M. K. Senan, M. Mustaqeem, and M. Sarfraz, "An intelligent and effective cyber-secured smart-home automation system with embedded AI," in *ICCAI 2021*, Springer, vol. 435, 2022, pp. 153–163. doi:10.1007/978-981-19-0976-4_13

### Patent

**P1** **M. K. Ikram**, "Measurement of real and reactive power using controlled rectifiers," Indian Patent Application No. 202111049900, published in the *Official Journal of the Patent Office*, Issue 46/2021, Nov. 2021.

## Teaching

**Sessional Lecturer** Aug 2025 – present

*Swinburne University of Technology, Melbourne, Australia*

- Design of Smart Power Grids (EEE80020 postgraduate; EEE40016 undergraduate): smart grid technologies, renewable integration and energy management. Developed course materials, lab exercises and assessments.
- Engineering Technologies with Indigenous Context (ENG10006): engineering from Aboriginal perspectives.

**MATLAB Trainer** Oct 2021 – May 2022

*Arena Multimedia, Aligarh, India*

- Trained more than 120 undergraduate students in MATLAB, Simulink and electrical engineering applications.

**Teaching Assistant** Feb 2019 – May 2020

*Aligarh Muslim University, India*

- Supervised instrumentation and control, electric machines and microprocessor labs for cohorts of up to 120 students.

## Awards and Recognition

**Cover article, World Electric Vehicle Journal, Vol. 17, Issue 8** 2026

**Siemens–Swinburne Energy Transition Hub PhD Scholarship (fully funded)** 2024 – 2027

**National GATE Scholarship, MHRD, Government of India** 2018 – 2020

GATE Electrical Engineering: 95th percentile, ranked 5,059 of 93,526 candidates.

## Service and Leadership

**Academic Senator (elected HDR student representative)** Jan 2026 – present

*Swinburne University of Technology*

- Represents postgraduate research students in university Senate and committee discussions on research and academic policy.

## Conference Presentations

**IEEE ETFG 2025, Wollongong, Australia** Dec 2025

**IEEE AUPEC 2025, Brisbane, Australia** 2025

**Energy Consortium, Melbourne, Australia** Nov 2024

**IEEE ICCCA 2021, Arad, Romania** Dec 2021

**Springer ICCAI 2021, Mathura, India** Nov 2021

## Professional Training

- PSS® E industry training, Modules 1 and 2, Siemens–Swinburne Energy Transition Hub
- PSS® SINCAL industry training, Siemens–Swinburne Energy Transition Hub
- Solar Rooftop Grid Engineer, National Institute of Solar Energy, India (2019)

## Skills

**Power system simulation:** PSS® E, PSS® SINCAL, MATLAB/Simulink

**Optimisation and automation:** Python (NumPy, Pandas, CVXPY), IBM ILOG CPLEX

**Hardware and testing:** Power electronics, sensor interfacing, analog circuits, electronic system design

**Data and documents:** Python (Matplotlib), Excel, LaTeX

**Languages:** English (full professional), Urdu and Hindi (native)

## Referees

Available on request.